

Lab 02 - Internet

Make a copy of this document for your work. Submit your work to the Lab 02 Assignment in D2L. Solutions will be posted the day after the lab is due and only for students who submit work. Check your answers - learning to self-assess is super important and will help you become a better computer scientist!

1. Explain, in your own words, what the four different layers of the internet (according to the TCP/IP model we learned in class) are and what they are responsible for?

2. Imagine that you're designing a protocol for playing a Wordle-clone. Each round you guess five letters and send it to the server and the server sends back information on whether the letters you guessed were correct and how many turns you have left.
 - a. How many bits are needed to send your guess to the server? Explain your reasoning.

- b. When the server responds it sends back the number of turns you have left (a number between 0 and 5) and, for each of the 5 letters, whether it is *completely wrong*, *wrong place but a correct letter*, or *completely correct* (note that is three possibilities). How many bits will the server send to you, the player? Explain your reasoning because there are multiple ways to do this correctly!